Session 3 — Project Tasks — P4A 2026

Complete these tasks in your group notebook before Session 4. Bring any questions to the start of Session 4.

What you covered today

- Loading data from CSV files and from a MySQL database
- Exploring a dataset: shape, column types, null values, statistics
- Filtering and selecting rows and columns
- GroupBy: aggregating data by category

Tasks

1. Load the staging table

Connect to the course database and load the `postings_with_benefits` table into a DataFrame. This table was created during the SQL sessions — it contains all active job postings (where `closed_time` is null) plus a `benefits` column with all listed benefits for each posting, concatenated and separated by `;`.

Confirm the shape and preview the first rows. Check that the `benefits` column is present.

2. Explore the benefits column

The `benefits` column contains a semicolon-separated string of benefits per posting — for example: `Medical insurance; 401K; Dental insurance`. Some postings have no benefits listed (null).

- What percentage of postings have at least one benefit listed?
- Split the column by `;`, expand the result, strip whitespace, and count the occurrences of each benefit. What are the top 10 most common benefits across all active postings?

Write one sentence on what this tells you about what companies typically offer on LinkedIn.

3. Load the companies data

Load the `companies_companies` table into a DataFrame. Confirm the shape and preview the first rows.

4. Assess data quality

For each of these columns, report the null percentage: company_size, country, city. Write a short markdown comment explaining what these null rates mean for your analysis.

5. Filter to US companies

Create a filtered DataFrame containing only companies where country == 'United States'. How many companies remain?

6. Count companies by size

Using the US companies, compute the number of companies for each value of company_size. The scale runs from 1 (smallest) to 7 (largest). Which size is most common? Which is least common?

7. Compute coverage for company size

Before drawing conclusions from Task 6, report **what percentage of US companies have a non-null company_size**. Write one sentence on what this means for any conclusions you draw from the size distribution.

8. Identify the top 10 countries

Using all companies (not just US), count the number of companies per country. List the top 10. Does the distribution surprise you?

Bring to Session 4

- Task 2: what are the top 3 most common benefits? Does this match what you would expect from a LinkedIn dataset?
- Task 4: which columns have too many nulls to be useful in your project analysis?
- Task 7: what percentage did you get for company_size coverage? Does that change how confident you are in the result?
- Task 8: the dataset is built from LinkedIn — does the country distribution reflect the actual global job market, or the platform's user base?